



france•tv

BBM SRv6 network



BBM : BackBone Media

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Agenda

1. **BBM Project context and objectives**
2. **Physical architecture – General topology**
3. **Routing foundations - SRv6 underlay**
4. **VPN services**
5. **Link Qualification**

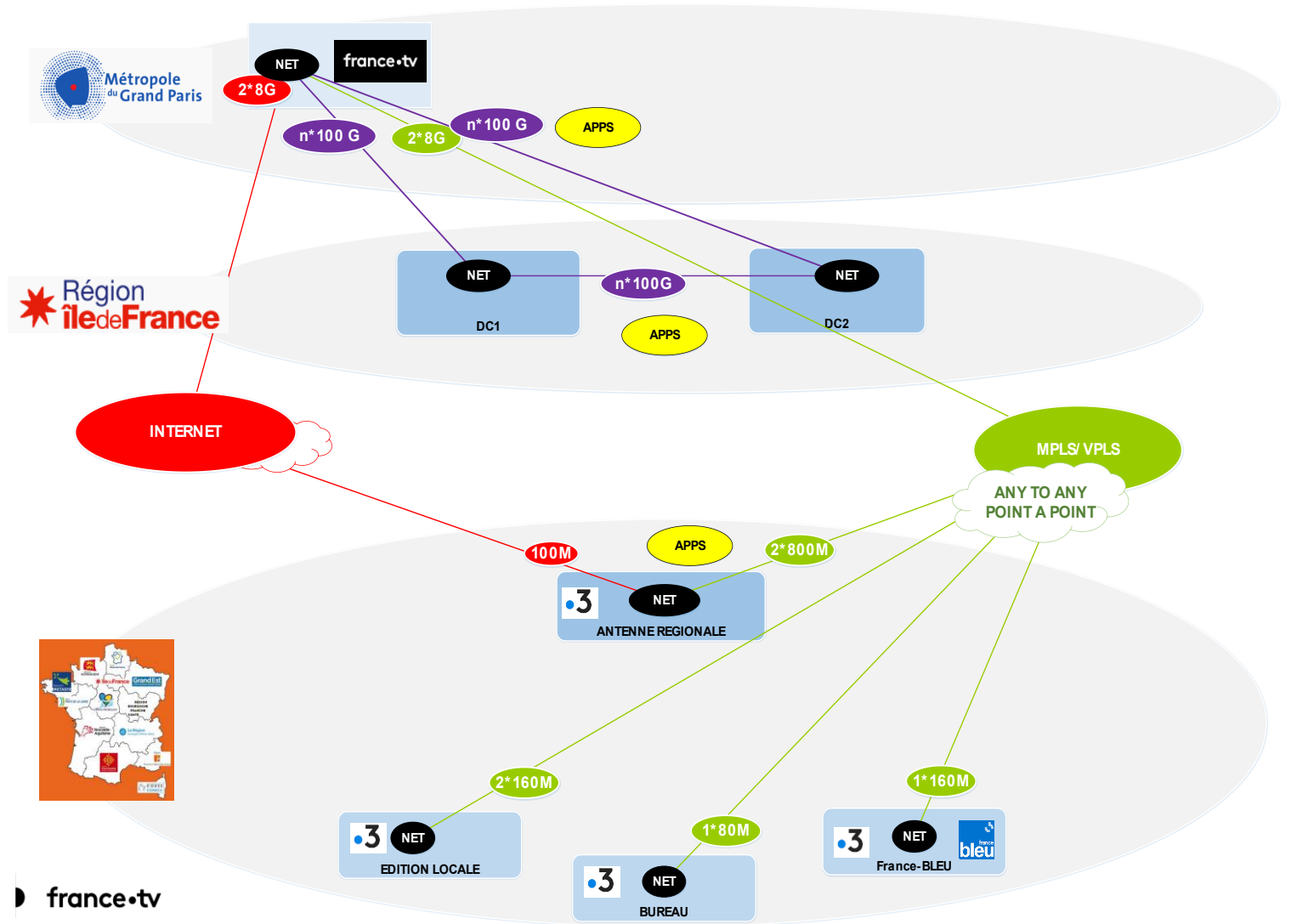
BBM Project

France Télévisions – A french TV Broadcaster

- French public broadcasting group, operating under the French State
- 5 national channels and multiple regional & overseas sites to interconnect
- A highly distributed organization with sites spread across metropolitan France and overseas territories
- About 9,000 users relying on a robust and available network infrastructure
- Critical broadcast & IT workflows requiring high availability and low-latency connectivity
- A complex multi-site WAN architecture supporting both administrative and production networks
- Growing need for hybrid connectivity to support cloud services and digital platforms (france.tv)

The legacy FranceTv WAN

- Network performance were limited to 10G.
- The applications were distributed locally in each regions
- The troubleshooting and the maintenance were done by the isp providers



BBM Project

So why FranceTv needed a new WAN network :

- Deploy the application centrally in the datacenter.
- Provide the needed resource to provide ST-2110 services on the wan.
- The consolidation of different team within the group
- Gain Flexibility to the services provided to the different IT teams within the group
- Enhance the Resources provided in the wan
- Have full control of the different communication through the wan without the need to go through the isp teams to troubleshoot any issue

BBM Project

France Télévisions designed BBM network to replace a legacy MPLS managed network.

This new backbone network interconnects:

- 2 Data Centers located in Paris area
- 26 Regional Branches located throughout the country
- ~50 remote sites attached to the regional branches

Key objectives:

- Ensure data exchange between headquarters, data centers, and the regional and remote sites
- Provide an extended service catalog (L3VPN, L2VPN, mVPN)
- Support ST2110 (SMPTE 2110) critical flow transport at high data rate
- Enable the hosting of other French public media players

The physical architecture relies on a 3 level-hierarchy

Level	Equipment	Throughput	Quantity
Core (DC)	Cisco 8804	100 Gb/s	2
Edge (RB)	Cisco 8011	100 Gb/s	52
Remote site	NCS540	1-10 Gb/s	~50

All routers are Cisco IOS XR devices.

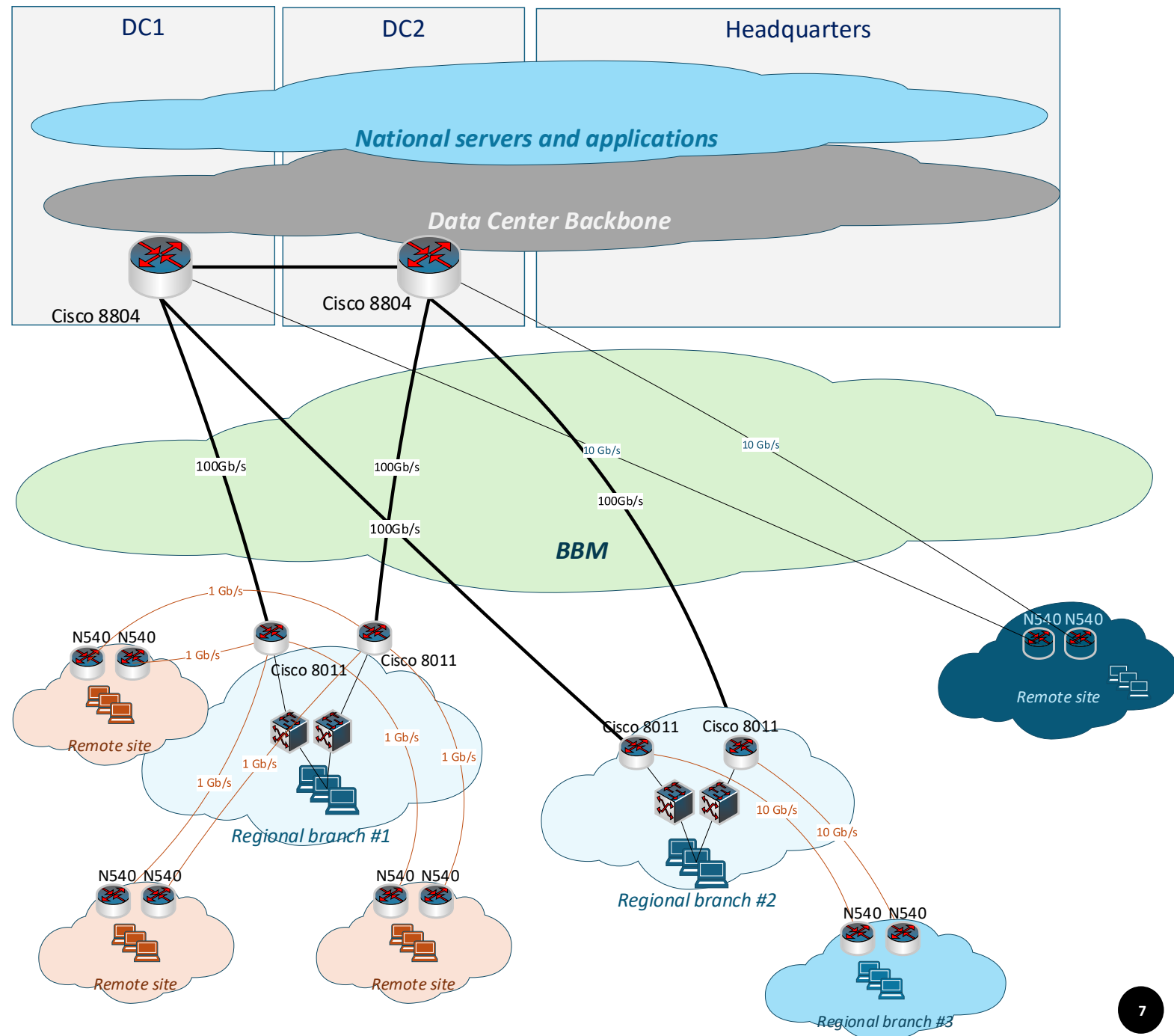
Interconnections are built on optical links provided by national operators

Data rate:

between DC and headquarters
➔ 4 x 100G breakouts

between DC and Regional branches
➔ 2 x 100G

between Regional branches and remote sites
➔ 1 to 10 G



Routing foundations

ISIS and SRv6 underlay

IPv6 instance for SRv6 transport
L2-only design
Flex Algos x3 (delay & red/blue disjoint planes)

Characteristics:

- Loopback address within each Regional SID Block
- WAN interfaces are IPv6 link-local only
- Micro-SID (uSID) in F3216 format

Addressing example for Site ID 9101:

Loopback0: fd01:0000:9101::1/128

SRv6 Locator: fd01:0000:9101::/48

IS-IS Net ID: 49.000.000.0000.9101.00

BGP overlay

Single ASN 4 bytes
Hierarchical Route Reflector design
IPv6 iBGP peering over Loopbacks

Address families activated:

- VPNv4 unicast
- L2VPN EVPN
- Link State
- MVPN (roadmap)

VPN services

L3VPN:

IPv4 unicast

Any to any

L2VPN:

Eline underlay for IPFM Fabrics

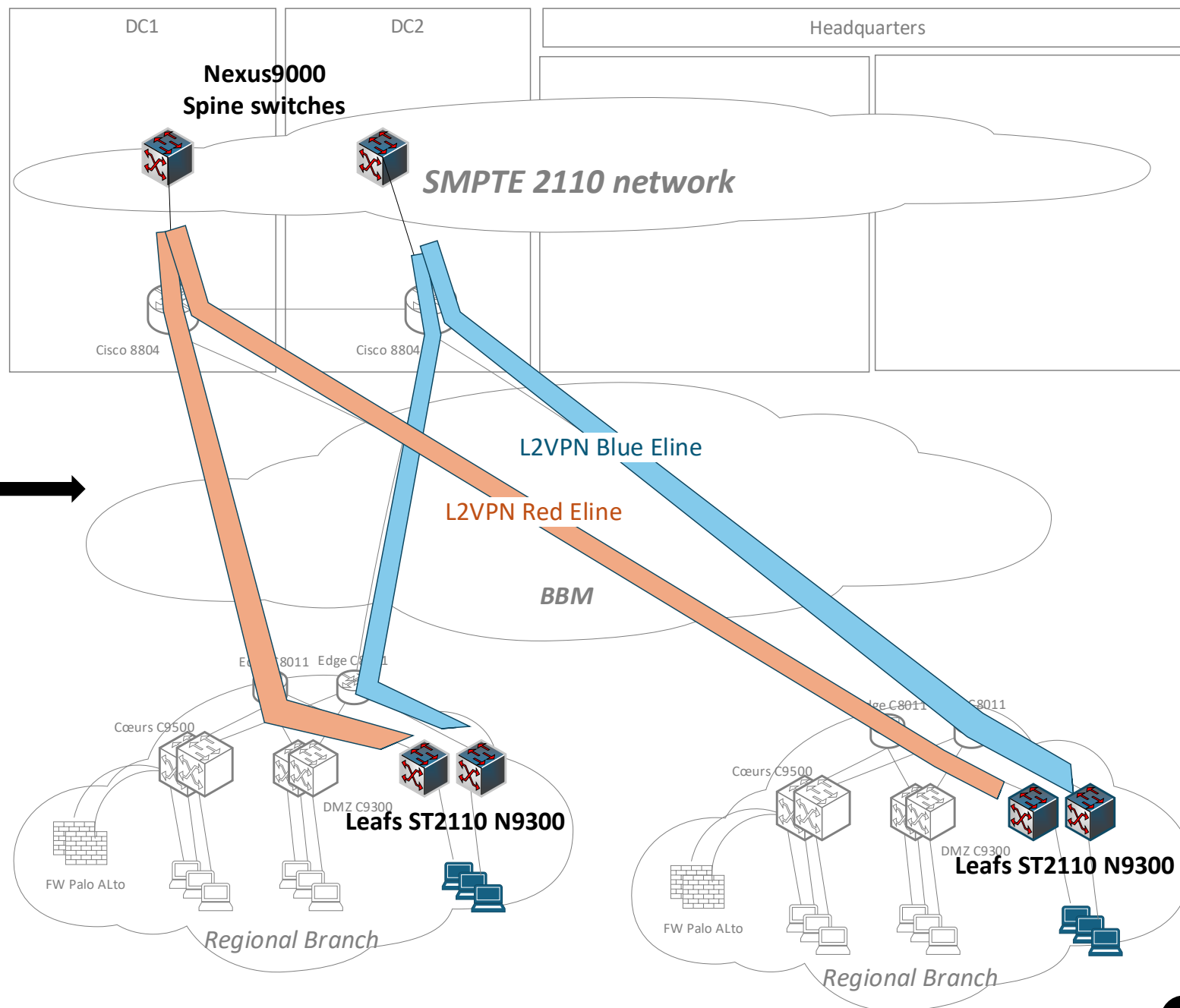
RED/BLE disjoint forwarding planes (SRv6 flex algos)

ST2110 traffic strict priority

MVPN:

BGP MVPN over IPv6 (Roadmap 2027)

IOS XR 26.4 Early Field Trial ongoing



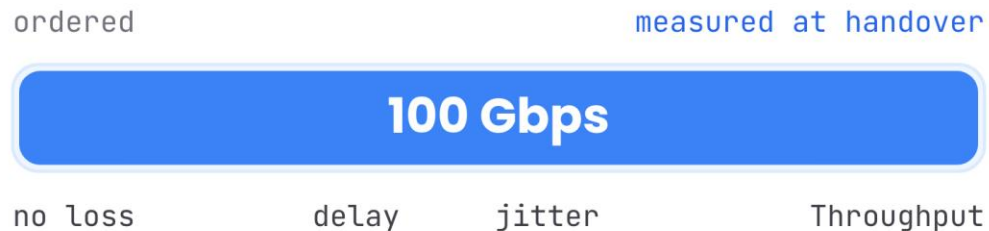
Automated link validation over SRv6

Link validation – The need

AT DELIVERY

A new 100G link is handed over.

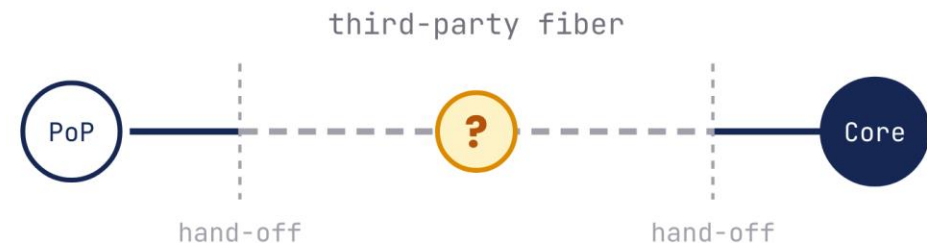
Qualify bandwidth, delay and jitter before it goes into service.
Fill it at line rate, and record the figures while it is full.



IN SERVICE

Throughput drops on a live link.

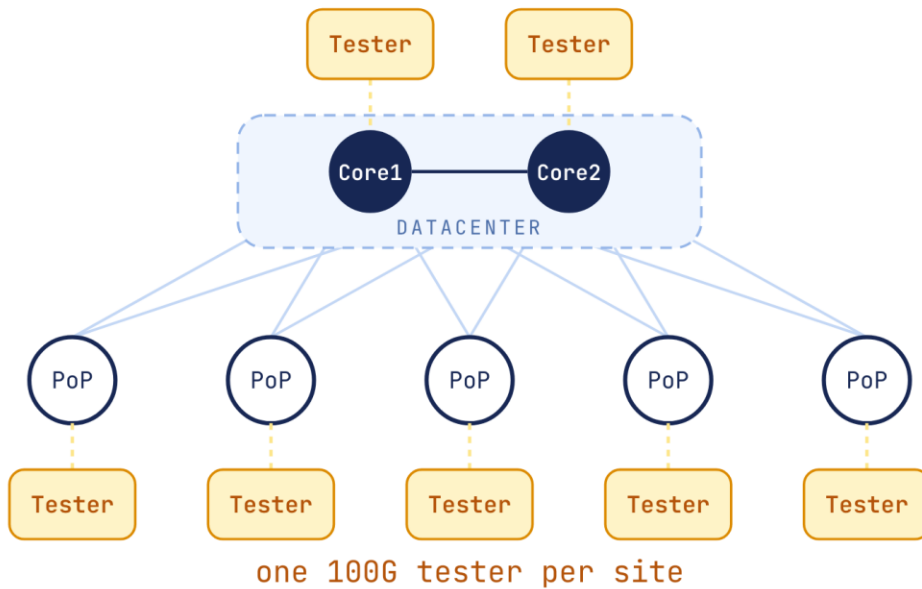
Re-qualify the link at 100G, or one QoS class on its own. The result says whether the fault sits in the fiber or in the France TV network, with figures the provider accepts.



Link validation – The options

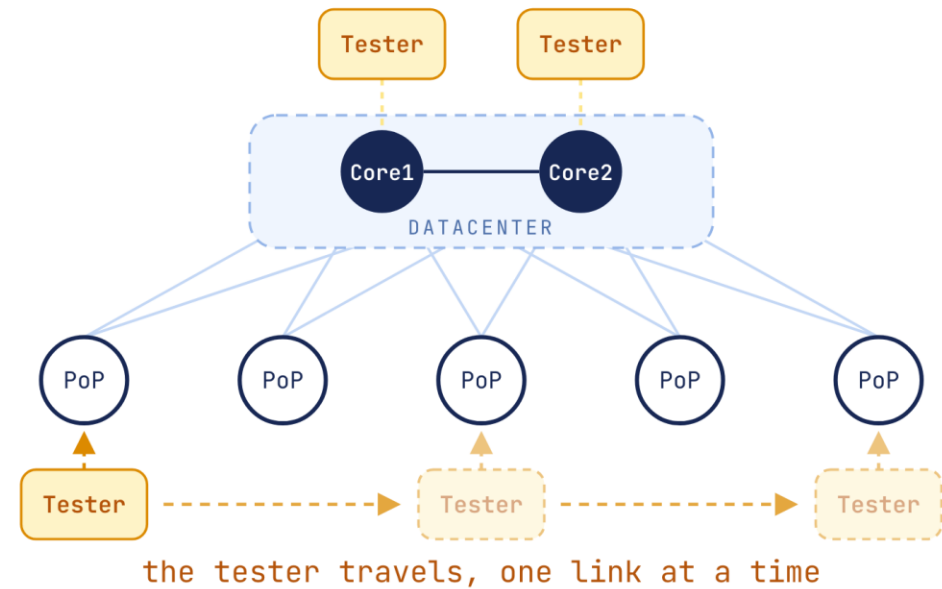
A tester at every site

TOO EXPENSIVE

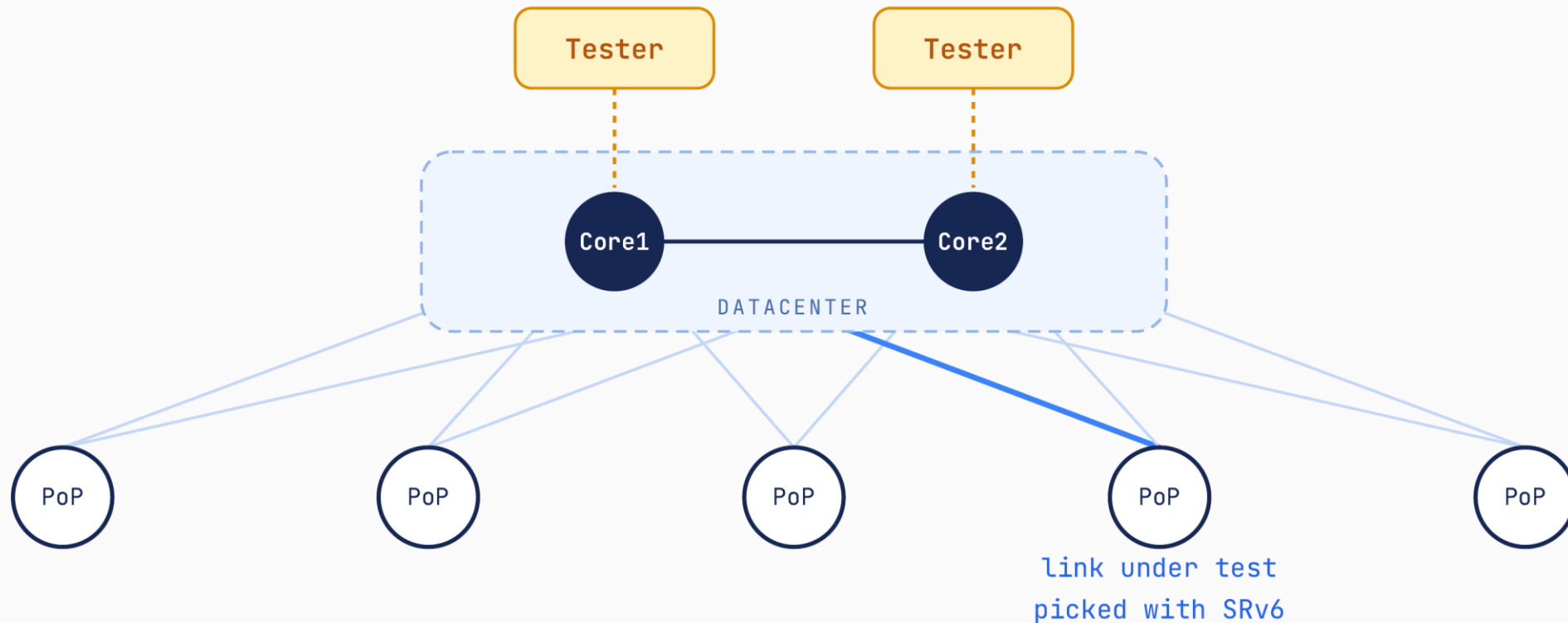


One tester that travels

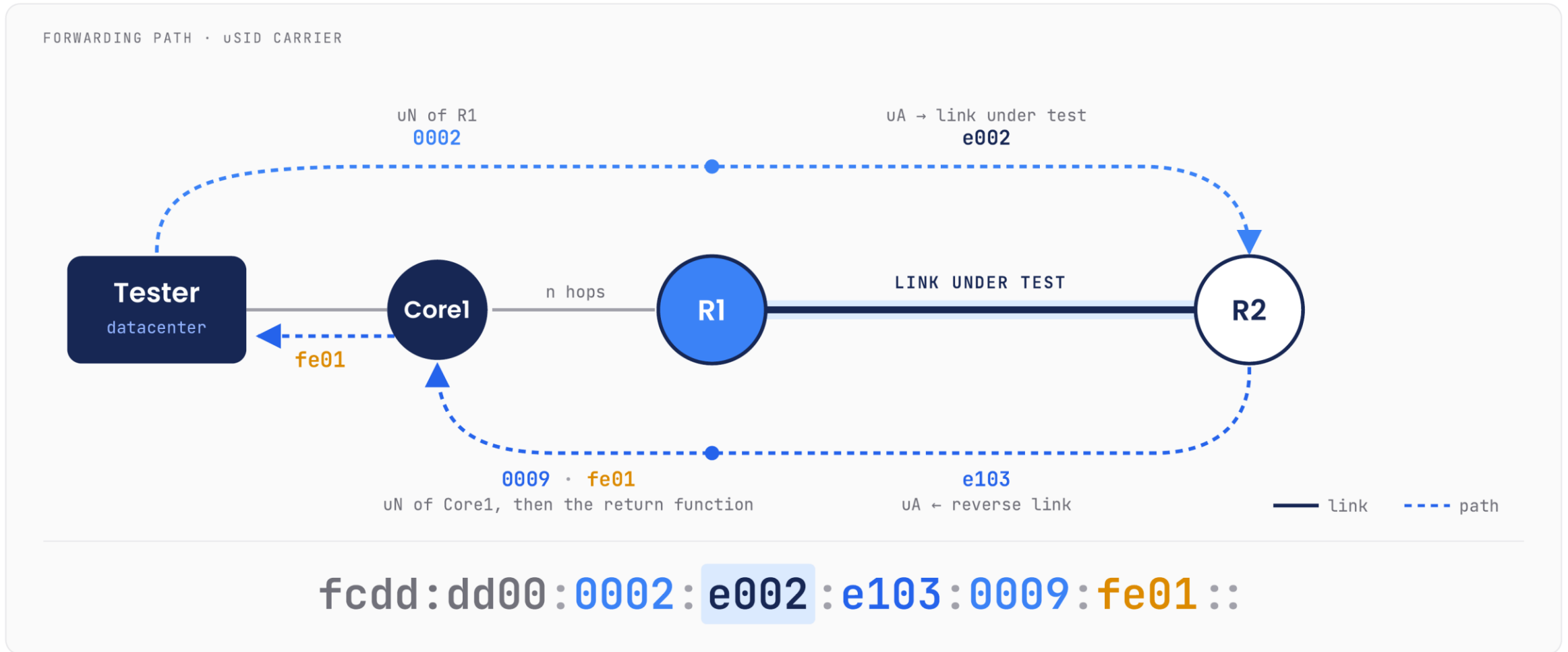
TOO SLOW



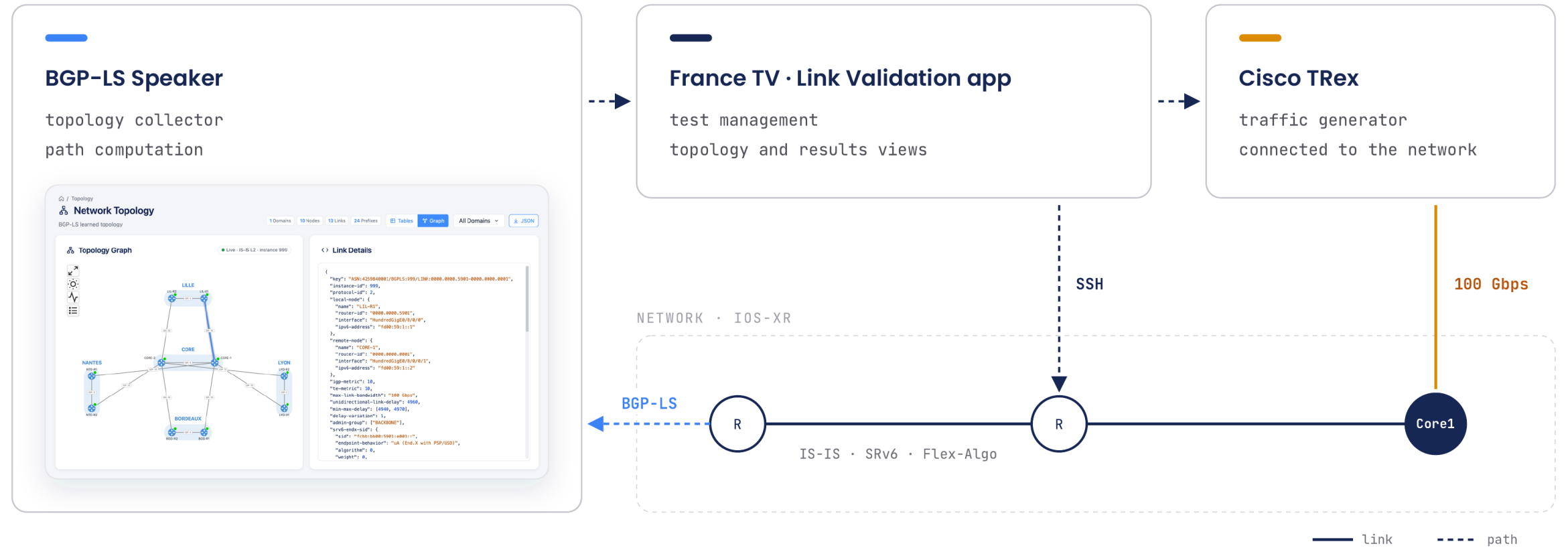
Link validation – The proposed solution



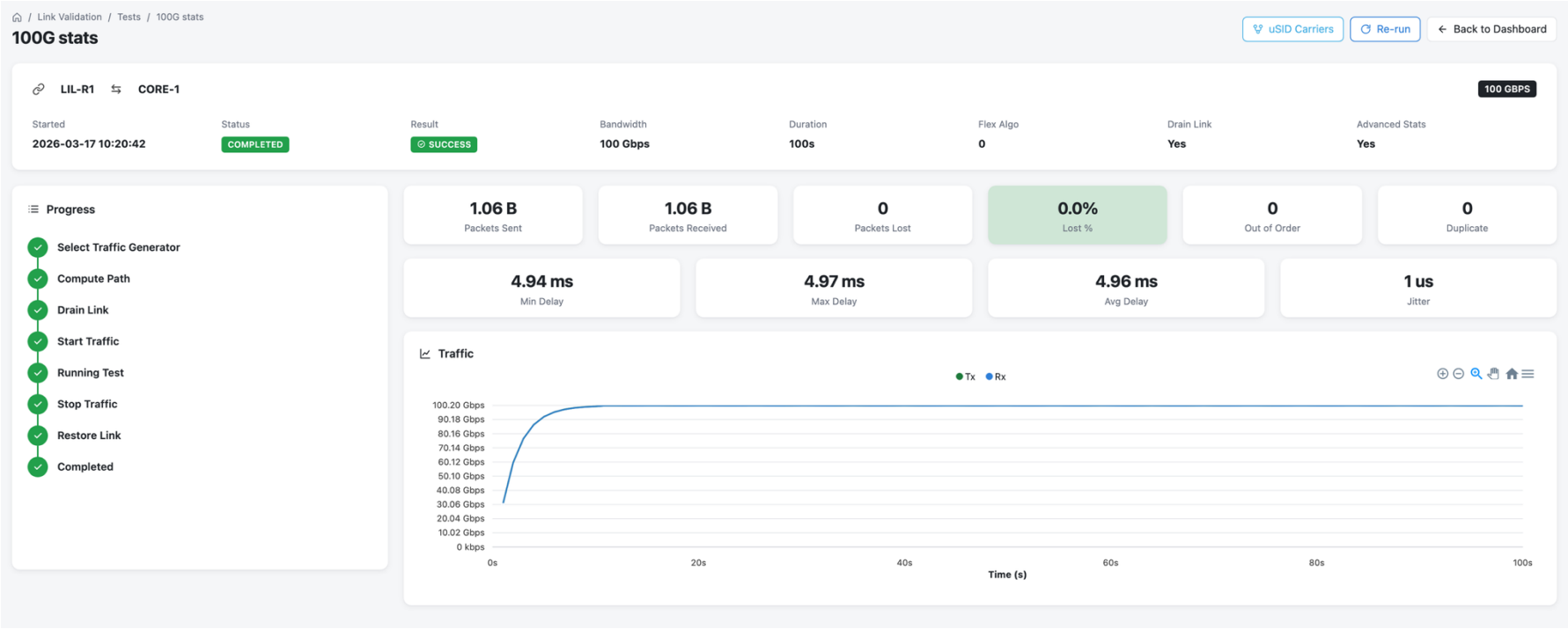
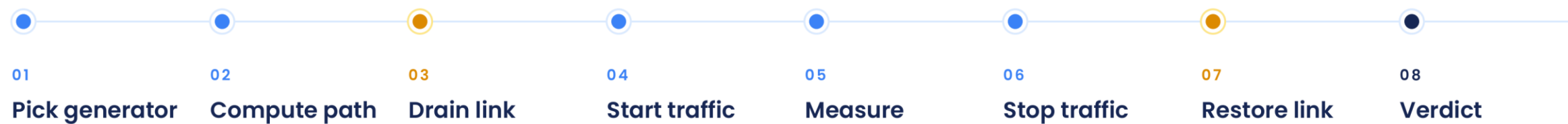
Link validation – The path as SRv6 uSID carrier



Link validation – Software stack



Link validation – One click, eight steps



Link validation - Key takeaways



01

No hardware per site

One tester reaches every link the IGP knows.



02

Data plane only

SRv6 sends the packets back on its own. The traffic stays in the data plane.



03

BGP-LS is the source of truth

Locators, SIDs, metrics, and the proof of convergence.



04

One tool, everything automated

One click, from picking the tester to the verdict. No CLI to type, no script to babysit.

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Thank you